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Subscription Scope Pattern

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Overview

The Subscription Scope Pattern demonstrates how to deploy resources at the Azure subscription level using the Release Engine framework. This pattern creates a Resource Group, which serves as a foundation for other resources and patterns.

Architecture

This pattern deploys a Resource Group at the subscription level, which can then host other Azure resources.

```
graph TB
    subgraph SUB ["Azure Subscription"]
        RG["Resource Group<br/>(example-avm-rg default)<br/><br/>(Ready for other resources)"]
    end

    style SUB fill:#fff3e0
    style RG fill:#ffcc80
```

Prerequisites

- **Azure Permissions:** Owner or Contributor access at subscription level
- **Service Principal:** Configured with subscription-level permissions
- **Subscription Access:** Active Azure subscription with sufficient quotas

Resources Created

- **Azure Resource Group** (AVM: avm/res/resources/resource-group:0.4.2)
 - Uses Azure Verified Modules for consistent naming and tagging
 - Configurable name and tags
 - Foundation for other resource deployments

Parameters

Parameter	Type	Description	Default	Required
tags	object	Tags to apply to the resource group	{}	No
resourceGroupName	string	Name of the resource group	example-avm-rg	No

Parameter Constraints

- **resourceGroupName:** Must follow Azure resource group naming conventions
- **tags:** Standard Azure tagging format (key-value pairs)

Outputs

Output	Type	Description
resourceGroupId	string	Resource ID of the created resource group

Deployment Scope

- **Target Scope:** Subscription
- **Service Connection Required:** Yes (subscription-level permissions)
- **Multi-Stage:** No (single deployment stage)

Pipeline Configuration

Stage Configuration

```
stages:
  - infrastructure:
      iac:
        name: subscription_deployment
        deploymentScope: Subscription
        serviceConnection: $(serviceConnection)
        iacMainFileName: subscription_scope_pattern.bicep
        iacParameterFileName: ${parameters.deploymentSettings.iacParameterFileName }
```

Environment Variables Required

The pattern expects these variables to be defined in your configuration repository:

```
# In vars-{environment}.yaml
variables:
  resourceGroupName: "rg-myapp-#{environmentAbbreviation}#"
  commonTags:
    Environment: " #{environmentAbbreviation}#"
    Owner: " #{primary_owner}#"
    CostCenter: " #{cost_center}#"
```

Configuration Repository Integration

Parameter File Example

(subscription_scope_pattern.parameters.json)

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentParameters.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "resourceGroupName": {
      "value": "rg-#{applicationName}-#{environmentAbbreviation}#"
    },
    "tags": {
      "value": {
        "Environment": " #{environmentAbbreviation}#",
        "Application": " #{applicationName}#",
        "Owner": " #{primary_owner}#",
        "CostCenter": " #{cost_center}#",
        "Workload": " #{workload}#"
      }
    }
  }
}
```

Metadata Configuration

```
# In _config/metadata.yaml
variables:
  workload: "subscription_scope_pattern" # Must match pattern name
  applicationName: "myapplication" # Used in resource group
naming
  primary_owner: "team@company.com" # For tagging
  cost_center: "IT-001" # For cost tracking
```

Usage Examples

Development Environment

```
# vars-development.yml
variables:
  environmentAbbreviation: dev
  resourceGroupName: "rg-myapp-dev"
  enableDiagnostics: true
```

Production Environment

```
# vars-production.yml
variables:
  environmentAbbreviation: prd
  resourceGroupName: "rg-myapp-prd"
  enableDiagnostics: true
  enableBackup: true
```

Security Considerations

- **Azure Verified Modules:** Uses official Microsoft-maintained modules
- **RBAC Ready:** Resource group created with proper structure for role assignments
- **Tagging Strategy:** Consistent tagging for governance and cost management
- **Audit Trail:** All operations logged at subscription level

Cost Optimization

- **Resource Group:** No direct cost for resource group creation
- **Tagging Strategy:** Enables cost allocation and tracking
- **Resource Policies:** Foundation for implementing cost governance policies
- **Budget Integration:** Resource group can be used for budget scoping

Monitoring and Diagnostics

The resource group supports:

- **Activity Logs:** Automatic logging of all resource group operations
- **Resource Health:** Monitoring of resources within the group
- **Policy Compliance:** Foundation for Azure Policy enforcement
- **Cost Analysis:** Grouping costs by resource group

Troubleshooting

Common Issues

Insufficient Permissions

Problem: Deployment fails with authorization errors

Solution: Ensure service principal has Contributor access at subscription level

Resource Group Already Exists

Problem: Resource group name conflicts

Solution: Use unique naming patterns with environment suffixes

Subscription Quota

Problem: Subscription limits reached

Solution: Check subscription quotas and request increases if needed

Validation Commands

```
# Validate Bicep template
az bicep build --file subscription_scope_pattern.bicep

# Test deployment (what-if)
az deployment sub what-if \
  --location westeurope \
  --template-file subscription_scope_pattern.bicep \
  --parameters resourceGroupName=test-rg
tags='{"Environment":"test"}'
```

Pattern Evolution

This pattern serves as a foundation for more complex patterns:

- **Multi-Resource Pattern:** Deploy multiple resources within the created resource group
- **Multi-Stage Pattern:** Create resource group first, then deploy dependent resources
- **Platform Pattern:** Create multiple resource groups for different tiers

Use Cases

Platform Foundation

- Creating standardized resource groups for platform services
- Establishing naming conventions and tagging strategies
- Preparing for multi-tier application deployments

Environment Setup

- Creating environment-specific resource groups
- Setting up development, test, and production boundaries
- Preparing for resource isolation and governance

Cost Management

- Establishing cost centers and billing boundaries
- Creating resource groups for different departments or projects
- Enabling detailed cost allocation and budgeting

Integration with Other Patterns

As Prerequisite

This pattern often serves as a prerequisite for:

- Single Resource Pattern (needs existing resource group)
- Application Patterns (foundation for app resources)
- Data Patterns (resource group for data services)

In Multi-Stage Deployments

```
# Example multi-stage with prerequisite
stages:
  # First stage: Create resource group
  - infrastructure:
      iac:
        name: foundation_stage
        deploymentScope: Subscription
        iacMainFileName: subscription_scope_pattern.bicep

  # Second stage: Deploy resources into the group
  - infrastructure:
      iac:
        name: resources_stage
        deploymentScope: ResourceGroup
        iacMainFileName: application_resources.bicep
        dependsOn: foundation_stage
```

Maintenance Notes

- **AVM Updates:** Monitor for updates to resource group AVM module
- **Tagging Evolution:** Update tagging strategy as governance requirements evolve
- **Naming Conventions:** Maintain consistency across all resource groups
- **Policy Alignment:** Ensure resource group creation aligns with Azure Policy requirements

Related Patterns

- **Single Resource Pattern:** Deploy resources into created resource group
- **Multi Stage Pattern:** Complex deployments using this as foundation
- **Platform Patterns:** Multiple resource groups for comprehensive platforms

This pattern provides the foundational subscription-level resource deployment capability and demonstrates proper use of Azure Verified Modules for governance and compliance.